

Placement Portal

Modern Application Development I — Project Report

A Flask-based campus recruitment management system

1. Student Details

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2. Project Details

Problem Statement:

To design and build a multi-role web application for managing campus recruitment. The platform provides a centralised system for posting placement drives, managing applications, and tracking candidate progress through the hiring pipeline — replacing manual, fragmented processes with a structured digital workflow.

Approach:

The application was built using Flask as the backend framework with an Application Factory pattern and Blueprint-based modular structure. It supports three distinct roles — Admin, Company, and Student — each with dedicated panels. Role-based access control, automated notifications, RESTful APIs, and resume handling were implemented and integrated end-to-end.

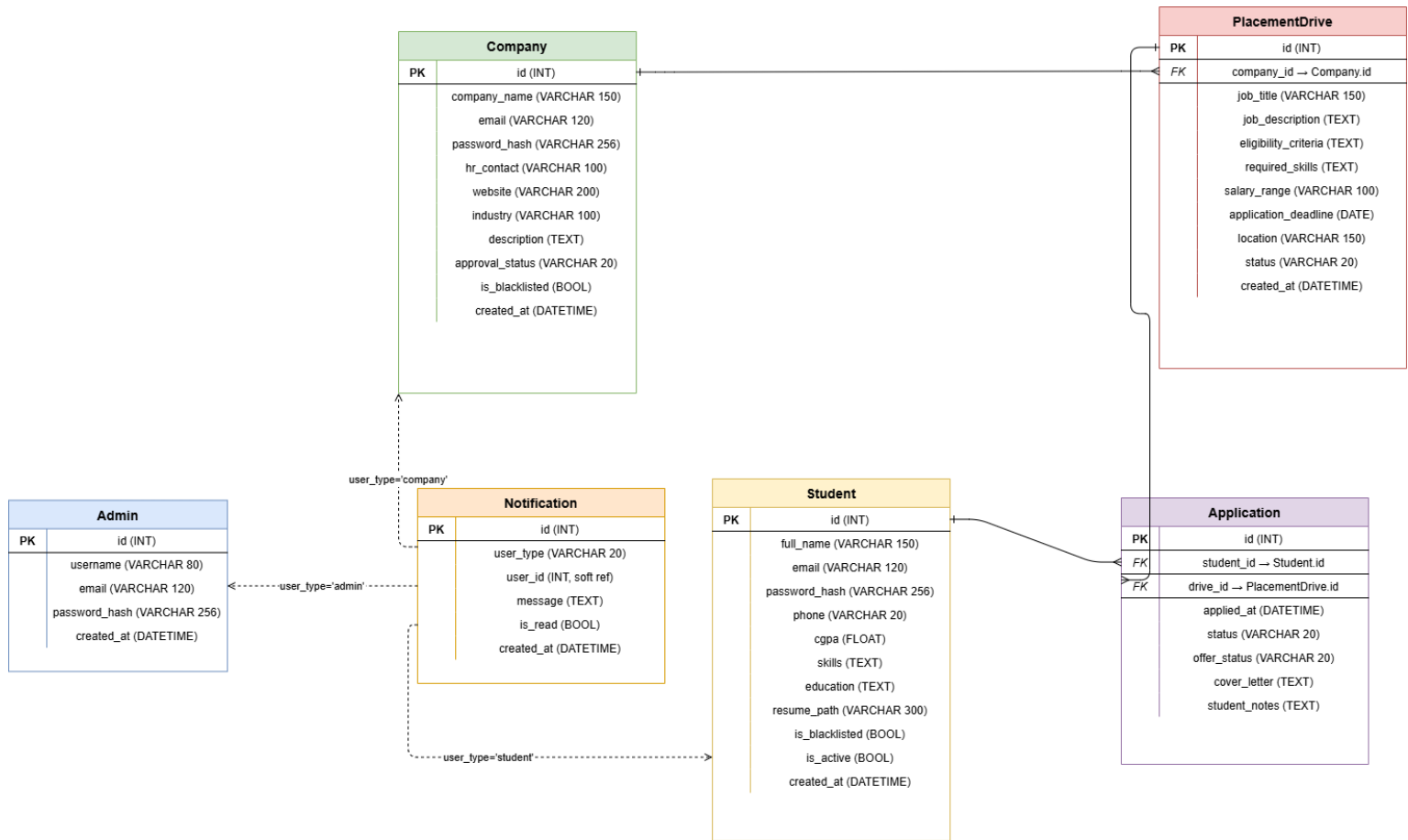
3. AI / LLM Usage Declaration

Claude (Anthropic) was used as a development assistant, primarily for debugging Jinja2 template logic, resolving Flask-RESTful request parser issues, and generating repetitive HTML/Bootstrap boilerplate for the admin CRUD views. All design, logic, debugging, and integration were done manually. **Estimated usage: ~20%**, limited to code suggestions and documentation formatting.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Python / Flask	Core web framework — lightweight, easy routing via Blueprints
Jinja2	Server-side HTML templating integrated with Flask
SQLite + SQLAlchemy	Relational database with ORM; SQLite chosen for zero-config local development
Flask-Login	Session management and role-based access control
Flask-RESTful	REST API layer for programmatic access to students and drives
Werkzeug	Password hashing (PBKDF2) and secure file upload utilities
Bootstrap 5	Responsive UI components; reduces custom CSS effort
Bootstrap Icons	Icon set consistent with Bootstrap 5 design language

5. Database Schema / ER Diagram



6. API Resource Endpoints

Endpoint	Methods	Description
/api/stats	GET	Returns aggregate counts — students, companies, drives, applications
/api/students	GET, POST	List all students; create a new student with full_name, email, phone, cgpa
/api/students/<id>	GET, PUT, DELETE	Retrieve, update, or delete a specific student by ID
/api/drives	GET, POST	List all drives; create a new drive with job_title and company_id
/api/drives/<id>	GET, PUT, DELETE	Retrieve drive; update its status (Approved/Rejected/Closed); delete drive

All endpoints accept and return JSON. Request parsing uses Flask-RESTful's RequestParser. The YAML (OpenAPI) definition is submitted as a separate file alongside this report.

7. Architecture and Features

Architecture Overview:

- app.py — defines create_app() using the Application Factory pattern; initialises extensions, registersBlueprints, sets up error handlers
- routes/auth — login, register, logout for all three user roles
- routes/admin — dashboard, student/company/drive management, blacklisting, bulk actions
- routes/company — drive CRUD, applicant management, status updates
- routes/student — browse drives, apply, profile management, notifications
- templates/ — Jinja2 templates mirroring Blueprint structure; shared base.html with navbar, flashmessages, Bootstrap CDN
- static/uploads/resumes/ — secure storage for student-uploaded resumes (PDF/DOC/DOCX)

Implemented Features:

Authentication & Authorisation

- Role-based login (Admin / Company / Student) via a single login form with radio selection
- Password hashing with Werkzeug PBKDF2; minimum 6-character enforcement on registration
- Flask-Login session management; per-Blueprint decorator guards (admin_required, company_required, student_required)
- Company accounts require admin approval before login; blacklisted accounts are blocked at login

Admin Panel

- Dashboard with real-time aggregate stats (total students, companies, drives, applications)
- Approve / Reject individual or bulk companies and drives
- Search and filter students and companies; blacklist / unblacklist any account
- View full student profile and application history; download/preview resume
- Embedded API request builder on the dashboard for live endpoint testing

Company Panel

- Post new placement drives (pending admin approval); edit, close, re-open, or delete drives
- View applicants per drive with tab-based status filters and CGPA sorting
- Update individual or bulk applicant status; automated notifications sent on every change• View student profiles and resumes restricted to applicants of own drives only

Student Panel

- Browse and search approved drives; apply with an optional cover letter (one application per driveenforced)
- Application history with status badges, offer acceptance/decline workflow, and personal notes
- Notification centre — unread notifications surfaced on dashboard, marked read on notification page• Profile management with resume upload (PDF/DOC/DOCX); resume viewable inline

8. Video Presentation

Drive Link: <https://drive.google.com/file/d/1g80tOs4j0xVdT5De7PpWF9nVWSRo4q3C/view?usp=sharing>
 (Accessible to all with 'View' permission.)